Relationships of water related leaf traits in the genus Anthurium

Project: PollObs (no title for now)

Provisional author position:

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Abstract

Flowering plants exhibit an astonishing diversity of reproductive structures, which shapes how they interact with their pollinators. However, to what extent reproductive traits or other drivers like phenology and species abundances shape plant-pollinator interactions is still

an open question. To disentangle the relative effects of species phenology, abundances and traits in plant-pollinator interactions, we compared pollinator visitation rates, diet breath, specialization from plant species that vary in reproductive traits, phenology and abundances. To do this, we observed plant-pollinator interactions for a whole flowering season in three different botanical gardens located in an urban setting. We anticipate that the ecological role of the different plan t and pollinator species are largely driven by their phenological overlap and abundance, and only certain taxonomic groups are strongly influenced by trait matching processes. Our results highlight the relevance of considering multiple mechanisms to understand plant-pollinator interactions.

Main objective

Explore how traits, phenology and abundances shape species interactions in three botanical gardens from Germany.

Pending tasks:

- Clean trait and phenology data for plants
- Get Pollinator phenology and traits
- Easy things to check: alpha, beta and gamma diversity of pollinators across the three gardens.

Methods

WRITE XXXXX

The data has an important temporal component

In Figure 1 we can see in a color coded NMDS the strong temporal component of the composition across gardens and different taxonomic levels.

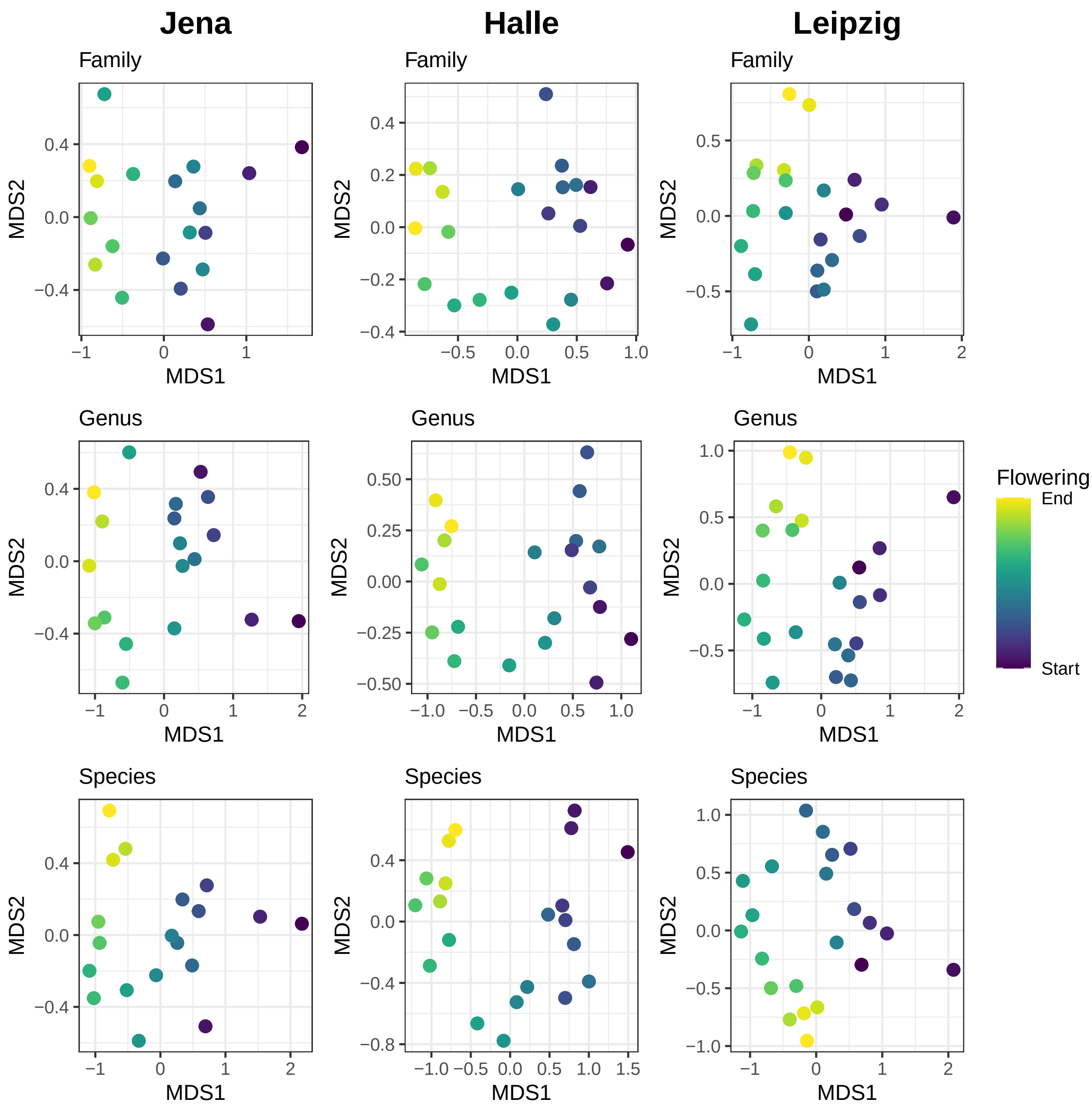

Figure 1. Non-metric multidimensional scaling (NMDS) of the pollinator community composition by sampling day in the three gardens at family, genus and species level.

Plant phenologies

The phenology of plants and pollinators was monitored weekly on each of the three gardens. Unlike plants, pollinator phenology relies on 'stochastic' interaction obervations that may not represent their complete flying period. Thus, we built a comprehensive flying period of all pollinators by retrieving GBIF records for all observed floral visitors during the sampling year. In addition, this download was limited by area, where we considered a rectangular shape that extends more on the longitude range than the latitude one, as we expect more homogeneous environmental conditions in the former (**Figure X**).

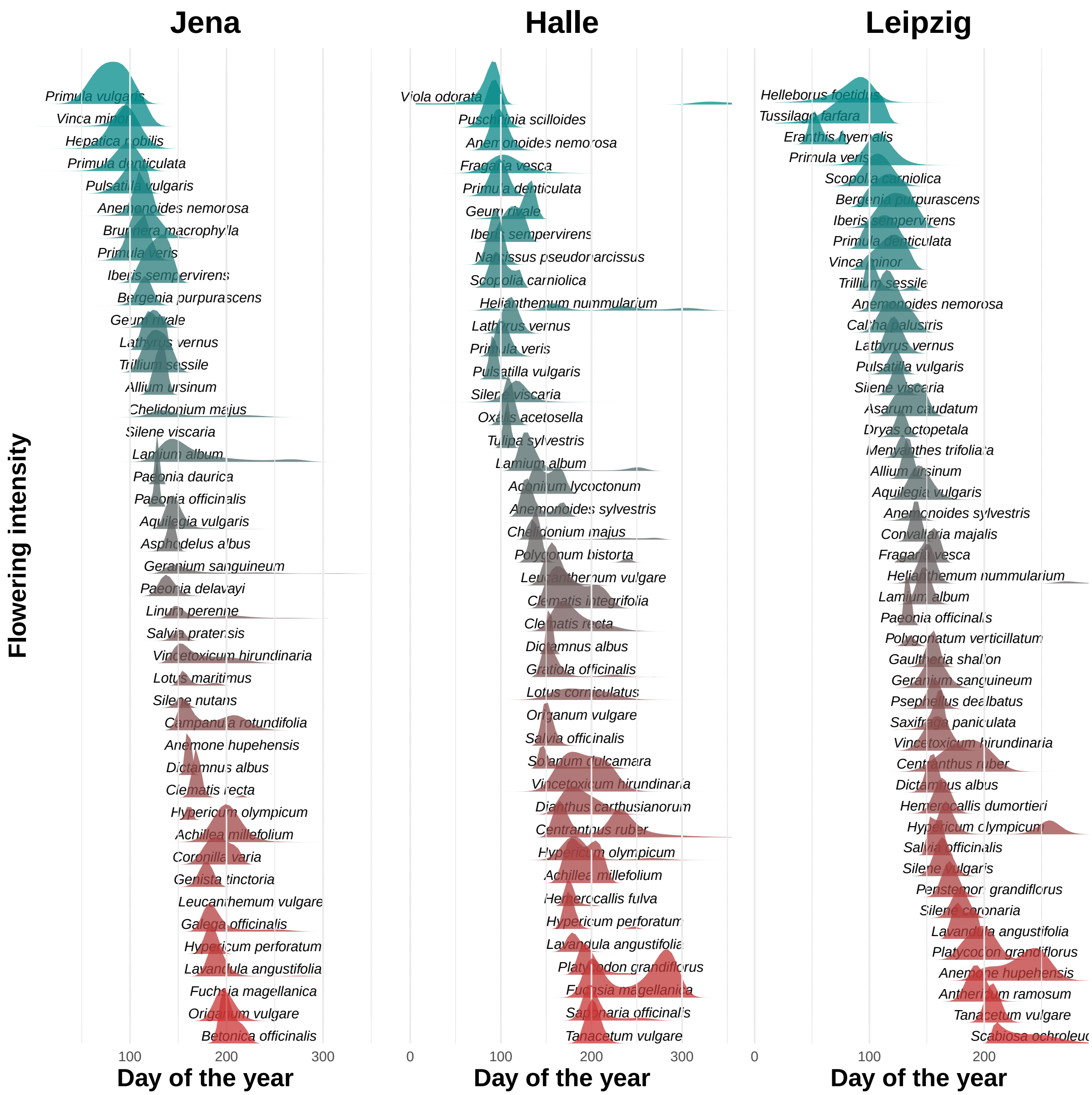
Jena
Halle
Leipzig
Flowering intensity
Day of the year
Day of the year
Day of the year
Primula vulgaris
Vinca minor
Hepatica nobilis
Primula denticulata
Pulsatilla vulgaris
Anemonoides nemorosa
Brunnera macrophylla
Primula veris
Iberis sempervirens
Bergenia purpurascens
Geum rivale
Lathyrus vernus
Trillium sessile
Allium ursinum
Chelidonium majus
Silene viscaria
Lamium album
Paeonia daurica
Paeonia officinalis
Aquilegia vulgaris
Asphodelus albus
Geranium sanguineum
Paeonia delavayi
Linum perenne
Salvia pratensis
Vincetoxicum hirundinaria
Lotus maritimus
Silene nutans
Campanula rotundifolia
Anemone hupehensis
Dictamnus albus
Clematis recta
Hypericum olympicum
Achillea millefolium
Coronilla varia
Genista tinctoria
Leucanthemum vulgare
Galega officinalis
Hypericum perforatum
Lavandula angustifolia
Fuchsia magellanica
Origanum vulgare
Betonica officinalis
Viola odorata
Puschkinia scilloides
Anemonoides nemorosa
Fragaria vesca
Primula denticulata
Geum rivale
Iberis sempervirens
Narcissus pseudonarcissus
Scopolia carniolica
Helianthemum nummularium
Lathyrus vernus
Primula veris
Pulsatilla vulgaris
Silene viscaria
Oxalis acetosella
Tulipa sylvestris
Lamium album
Aconitum lycoctonum
Anemonoides sylvestris
Chelidonium majus
Polygonum bistorta
Leucanthemum vulgare
Clematis integrifolia
Clematis recta
Dictamnus albus
Gratiola officinalis
Lotus corniculatus
Origanum vulgare
Salvia officinalis
Solanum dulcamara
Vincetoxicum hirundinaria
Dianthus carthusianorum
Centranthus ruber
Hypericum olympicum
Achillea millefolium
Hemerocallis fulva
Hypericum perforatum
Lavandula angustifolia
Platycodon grandiflorus
Fuchsia magellanica
Saponaria officinalis
Tanacetum vulgare
Helleborus foetidus
Tussilago farfara
Eranthis hyemalis
Primula veris
Scopolia carniolica
Bergenia purpurascens
Iberis sempervirens
Primula denticulata
Vinca minor
Trillium sessile
Anemonoides nemorosa
Caltha palustris
Lathyrus vernus
Pulsatilla vulgaris
Silene viscaria
Asarum caudatum
Dryas octopetala
Menyanthes trifoliata
Allium ursinum
Aquilegia vulgaris
Anemonoides sylvestris
Convallaria majalis
Fragaria vesca
Helianthemum nummularium
Lamium album
Paeonia officinalis
Polygonatum verticillatum
Gaultheria shallon
Geranium sanguineum
Psephellus dealbatus
Saxifraga paniculata
Vincetoxicum hirundinaria
Centranthus ruber
Dictamnus albus
Hemerocallis dumortieri
Hypericum olympicum
Salvia officinalis
Silene vulgaris
Penstemon grandiflorus
Silene coronaria
Lavandula angustifolia
Platycodon grandiflorus
Anemone hupehensis
Anthericum ramosum
Tanacetum vulgare
100
200
300
0
100
200
300
0
100
200

Figure 2.